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MECHANICAL VIBRATIONS L4

FREE VIBRATION OF 2 DOF UNDAMPED SYSTEMS

CONTENTS

- Free vibration of Undamped 2 DOF
- Equation of motion
- Mode shape of 2 DOF system
- Response of 2 DOF system
- Mode Shape plot

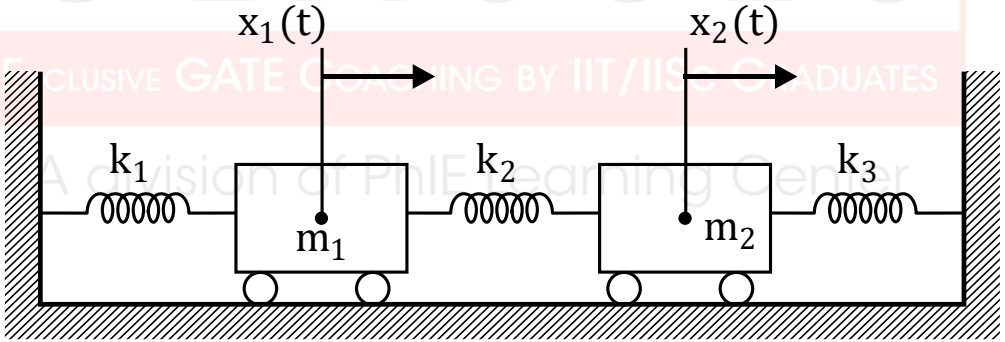
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Free Vibration of Undamped 2 DOF

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The system which requires two coordinates independently to describe its motion completely is called a two degree of freedom system.

Let two masses m_1 and m_2 vibration system oscillating on paper plane as shown in fig.



PROBLEM 1

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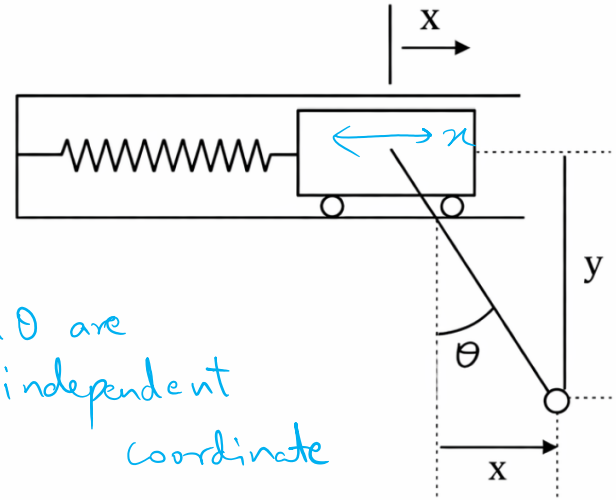
A mass is attached to a spring and placed horizontally in a frictionless surface. A simple pendulum has been pivoted to the mass. The degree of freedom of this system is

(A) 1

(B) 2

(C) 3

(D) 4



r & θ are independent coordinate

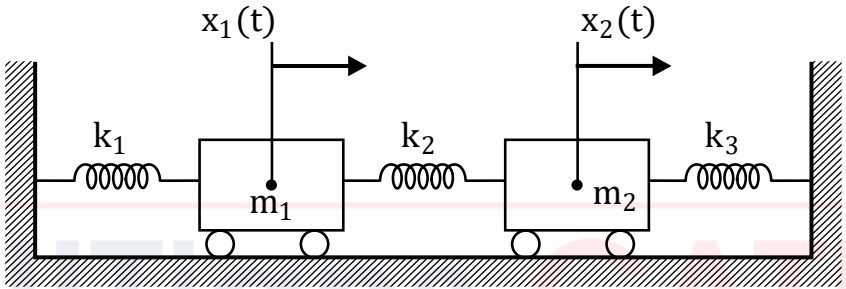
Ans b

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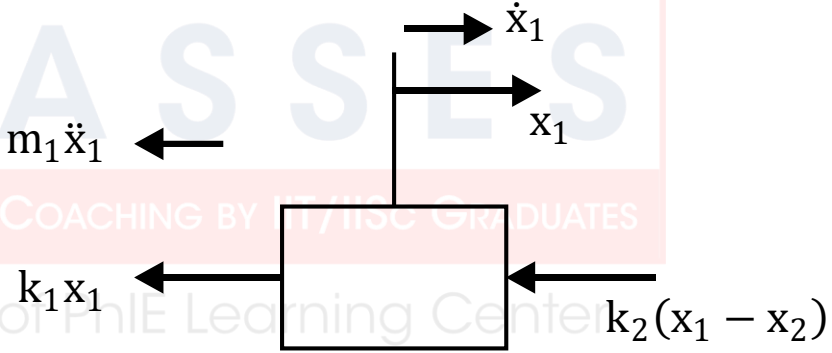
EQUATION OF MOTION

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Let $x_1 > x_2$

Equation of motion for m_1



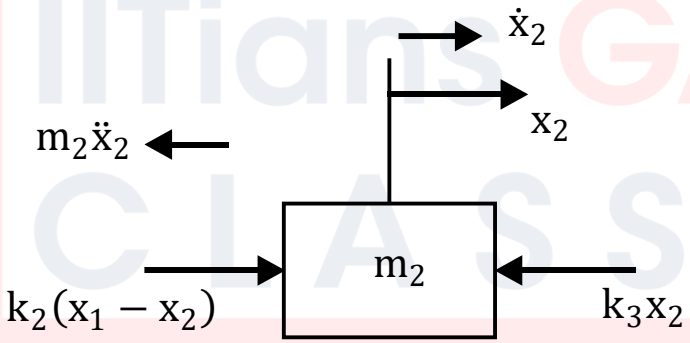
$$m_1\ddot{x}_1 + k_1x_1 + k_2(x_1-x_2) = 0$$

$$m_1\ddot{x}_1 + (k_1+k_2)x_1 - k_2x_2 = 0 \quad \text{..... (1)}$$

EQUATION OF MOTION

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Equation of motion for m_2

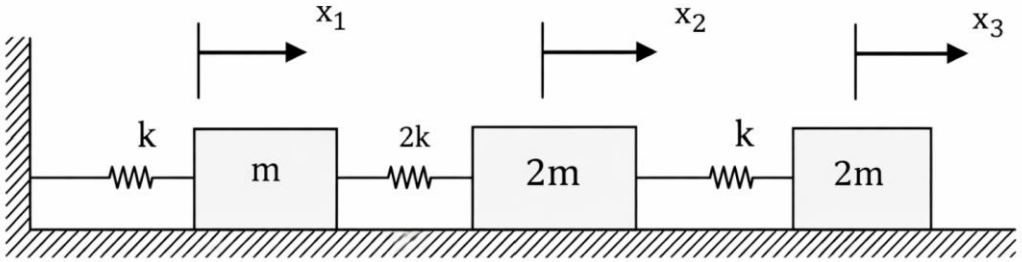


$$m_2\ddot{x}_2 + k_3x_2 + k_2(x_2 - x_1) = 0$$

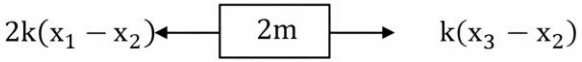
$$m_2\ddot{x}_2 + (k_3 + k_2)x_2 - k_2x_1 = 0 \quad \text{..... (2)}$$

PROBLEM 2

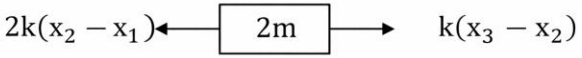
Find out the correct free body diagram for the middle mass for the given spring-mass system in below figure



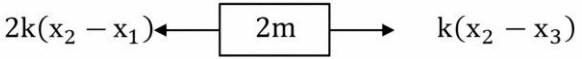
(A)



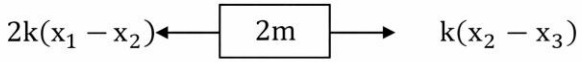
(B)



(C)



(D)



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Ans d

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EQUATION OF MOTION

$$m_1\ddot{x}_1 + (k_1+k_2)x_1 - k_2x_2 = 0 \quad \text{..... (1)}$$

$$m_2\ddot{x}_2 + (k_3+k_2)x_2 - k_2x_1 = 0 \quad \text{..... (2)}$$

$m\ddot{x} + kx = 0 \longrightarrow [M]_{1\times1}\ddot{x} + [k]_{1\times1}x = [0]$

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_3+k_2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Mass Matrix
↳ Symmetric
↳ Diagonalized

Stiffness Matrix
↳ Symmetric

$|\text{Symmetric matrix}| = 0$

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EQUATION OF MOTION

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Let, $x_1 = X_1 \cos(\omega t + \phi)$; $x_2 = X_2 \cos(\omega t + \phi)$

$\ddot{x}_1 = -X_1 \omega^2 \cos(\omega t + \phi)$ $\quad \ddot{x}_2 = -\omega^2 X_2 \cos(\omega t + \phi)$

Then $\ddot{x}_1 = -\omega^2 x_1$ $\quad \ddot{x}_2 = -\omega^2 x_2$

$$[\{-m_1\omega^2 + (k_1 + k_2)\}X_1 - k_2X_2] \cos(\omega t + \phi) = 0$$

$$[-k_2X_1 + \{-m_2\omega^2 + (k_2 + k_3)\}X_2] \cos(\omega t + \phi) = 0$$

$$\{-m_1\omega^2 + (k_1 + k_2)\}X_1 - k_2X_2 = 0 \quad \ldots\ldots (3)$$

$$-k_2X_1 + \{-m_2\omega^2 + (k_2 + k_3)\}X_2 = 0 \quad \ldots\ldots (4)$$

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$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_3+k_2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{bmatrix} -\omega^2 x_1 \\ -\omega^2 x_2 \end{bmatrix} + \begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_3+k_2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left\{\begin{bmatrix} -m_1\omega^2 & 0 \\ 0 & -m_2\omega^2 \end{bmatrix} + \begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_3+k_2 \end{bmatrix}\right\}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -m_1\omega^2 + k_1+k_2 & -k_2 \\ -k_2 & -m_2\omega^2 + k_3+k_2 \end{bmatrix}\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}\cos(\omega t+\phi) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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EQUATION OF MOTION

$\omega = \sqrt{\frac{k}{m}}$

$m\ddot{x} + kx = 0$

$x = A\cos(\omega t)$

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$$\begin{bmatrix} -m_1\omega^2 + (k_1 + k_2) & -k_2 \\ -k_2 & -m_2\omega^2 + (k_2 + k_3) \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$x_1 = X_1\cos(\omega t)$

$x_2 = X_2\cos(\omega t)$

natural frequencies

This represents two simultaneous homogenous algebraic equations in the unknown X_1 and X_2. It is satisfied by the trivial solution $X_1 = X_2 = 0$ which implies there is no vibration.

For a non-trivial solution of X_1 and X_2, the determinant of the coefficient of X_1 and X_2 must be zero, because X_1 and X_2 are dependent & ω is unkown

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$$\det\begin{bmatrix} -m_1\omega^2 + (k_1 + k_2) & -k_2 \\ -k_2 & -m_2\omega^2 + (k_2 + k_3) \end{bmatrix} = 0$$

$\Rightarrow$ $\omega \longrightarrow$ values

EQUATION OF MOTION

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$$(m_1m_2)\omega^4 - \{(k_1 + k_2)m_2 + (k_2+k_3)\, m_1\}\omega^2 + \{(k_1 + k_2)(k_2 + k_3) - {k_2}^2\} = 0$$

$$(m_1m_2)\omega^4 - \{m_1(k_3 + k_2) + m_2(k_1+k_2)\}\omega^2 + \{k_1k_2 + k_2k_3 + k_3k_1\} = 0$$

$${\omega_1}^2, {\omega_2}^2$$

$$= \frac{1}{2}\left\{\frac{(k_1 + k_2)\, m_2 + (k_2+k_3)\, m_1}{m_1m_2}\right\}$$

$$\pm \frac{1}{2}\left[\left\{\frac{(k_1 + k_2)\, m_2 + (k_2+k_3)\, m_1}{m_1m_2}\right\}^2 - 4\times1\left\{\frac{(k_1 + k_2)\,(k_2 + k_3) - {k_2}^2}{m_1m_2}\right\}\right]^{1/2}$$

This shows that it is possible for the system to have a positive nontrivial harmonic solution when ω is equal to ω_1 and ω_2. Here ω_1 and ω_2 is natural frequency of the system.

ω_1 & ω_2

EQUATION OF MOTION

If suppose we know modes (i.e ω_1 & ω_2)

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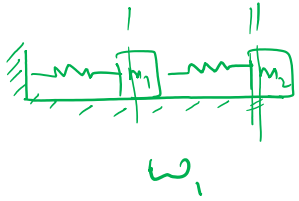
Here determinant is homogenous, only ratio of amplitude of frequencies can be found.

$\omega_1 \rightarrow$ Mode shape

$$1 \quad r_1 = \frac{X_2^{(1)}}{X_1^{(1)}} = \frac{-m_1\omega_1^2 + (k_1 + k_2)}{k_2} = \frac{k_2}{-m_2\omega_1^2 + (k_2 + k_3)} \; ;$$

(1st row) (2nd row)

(r_1 = amplitude ratio at frequency ω_1)



$$r_2 = \frac{X_2^{(2)}}{X_1^{(2)}} = \frac{-m_1\omega_2^2 + (k_1 + k_2)}{k_2} = \frac{k_2}{-m_2\omega_2^2 + (k_2 + k_3)} \; ;$$

(r_2 = amplitude ratio at frequency ω_2)

Amplitude → frequency/mode × particle number

MODE SHAPE OF 2 DOF SYSTEM

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The normal modes of vibration corresponding to ω_1 and ω_2 can be expressed.

Mode shape 1 $\leftarrow \omega_1$ Mode shape 2 $\leftarrow \omega_2$

$$\vec{X}^{(1)} = \begin{Bmatrix} X_1^{(1)} \\ X_2^{(1)} \end{Bmatrix} = \begin{Bmatrix} X_1^{(1)} \\ r_1 X_1^{(1)} \end{Bmatrix} \quad \& \quad \vec{X}^{(2)} = \begin{Bmatrix} X_1^{(2)} \\ X_2^{(2)} \end{Bmatrix} = \begin{Bmatrix} X_1^{(2)} \\ r_2 X_1^{(2)} \end{Bmatrix}$$

The vectors $\vec{X}^{(1)}$ and $\vec{X}^{(2)}$, which denote the normal mode of vibration are known as 'mode vectors' of the system.

EOM $= 0$

1st particle, 2nd particle

$$\vec{x}^{(1)} = \begin{Bmatrix} x_1^{(1)}(t) \\ x_2^{(1)}(t) \end{Bmatrix} = \begin{Bmatrix} X_1^{(1)} \cos(\omega_1 t + \phi_1) \\ r_1 X_1^{(1)} \cos(\omega_1 t + \phi_1) \end{Bmatrix}$$

single row solved.

first mode

$$\vec{x}^{(2)} = \begin{Bmatrix} x_1^{(2)}(t) \\ x_2^{(2)}(t) \end{Bmatrix} = \begin{Bmatrix} X_1^{(2)} \cos(\omega_2 t + \phi_2) \\ r_2 X_1^{(2)} \cos(\omega_2 t + \phi_2) \end{Bmatrix}$$

second mode

$X_1^{(1)}, X_1^{(2)}, \phi_1$ and ϕ_2 are determined by initial condition

RESPONSE OF 2 DOF SYSTEM

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Initial Condition

For mass m_1 and m_2 initial condition is given.

$x_1(t=0) = x_1(0)$; $\dot{x}_1(t=0) = \dot{x}_1(0)$

$x_2(t=0) = x_2(0)$; $\dot{x}_2(t=0) = \dot{x}_2(0)$

$x_1^{(1)}(t)$ ← displ. of particle m_1 at mode 1

$x_1^{(2)}(t)$ ← displ. of particle m_1 at mode 2

Motion for mass m_1 can be written as

$$x_1(t) = x_1^{(1)}(t) + x_1^{(2)}(t) \quad ; \text{(motion with } \omega_1 \text{ and } \omega_2\text{)}$$

← Response of particle m_1

By concept of Linear combination

$$x_1(t) = X_1^{(1)} \cos(\omega_1 t + \phi_1) + X_1^{(2)} \cos(\omega_2 t + \phi_2) \qquad \text{- (1)}$$

$$\dot{x}_1(t) = -\omega_1 X_1^{(1)} \sin(\omega_1 t + \phi_1) - \omega_2 X_1^{(2)} \sin(\omega_2 t + \phi_2) \qquad \text{- (2)}$$

4DOF $x_1 = x_1^{(1)}(t) + x_1^{(2)}(t) + x_1^{(3)}(t) + x_1^{(4)}(t)$

RESPONSE OF 2 DOF SYSTEM

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Motion for mass m_2 can be written as

$$x_2(t) = x_2^{(1)}(t) + x_2^{(2)}(t) \quad ;(\text{ motion with } \omega_1 \text{ and } \omega_2)$$

$$x_2(t) = X_2^{(1)} cos(\omega_1 t + \phi_1) + X_2^{(2)} cos(\omega_2 t + \phi_2)$$

$$x_2(t) = r_1 X_1^{(1)} cos(\omega_1 t + \phi_1) + r_2 X_1^{(2)} cos(\omega_2 t + \phi_2) \quad - (3)$$

$$\dot{x_2}(t) = -\omega_1 r_1 X_1^{(1)} sin(\omega_1 t + \phi_1) - \omega_2 r_2 X_1^{(2)} sin(\omega_2 t + \phi_2) \quad - (4)$$

Putting the four given initial condition in $x_1(t)$, $\dot{x}_1(t)$, $x_2(t)$ and $\dot{x}_2(t)$ to obtain $X_1^{(1)}$, $X_1^{(2)}$, ϕ_1 and ϕ_2.

r_1 and r_2 are already known from amplitude ratio derived. So $X_2^{(1)}$ and $X_2^{(2)}$ can be find.

4 DOF

$$\vec{x}^{(1)} = \begin{Bmatrix} x_1^{(1)}(t) \\ x_2^{(1)}(t) \end{Bmatrix} = \begin{Bmatrix} X_1^{(1)} \cos(\omega_1 t + \phi_1) \\ r_1 X_1^{(1)} \cos(\omega_1 t + \phi_1) \end{Bmatrix} \qquad \vec{x}^{(2)} = \begin{Bmatrix} x_1^{(2)}(t) \\ x_2^{(2)}(t) \end{Bmatrix} = \begin{Bmatrix} X_1^{(2)} \cos(\omega_2 t + \phi_2) \\ r_2 X_1^{(2)} \cos(\omega_2 t + \phi_2) \end{Bmatrix}$$

$x_3^{(1)}(t)$

$x_4^{(1)}(t)$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} = \begin{bmatrix} x_1^{(1)}(t) + x_1^{(2)}(t) + x_1^{(3)}(t) + x_1^{(4)}(t) \\ x_2^{(1)}(t) + x_2^{(2)}(t) + x_2^{(3)}(t) + x_2^{(4)}(t) \\ \vdots \\ x_4^{(1)}(t) + x_4^{(3)}(t) + x_4^{(4)}(t) \end{bmatrix}$$

Rows → particle

Columns → Mode shape

Response of each particle

Mode shape 1

Mode shape 2

Mode shape 3

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RESPONSE OF 2 DOF SYSTEM

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$$X_1^{(1)} = \frac{1}{(r_2 - r_1)}\left[\{r_2 x_1(0) - x_2(0)\}^2 + \frac{\{-r_2\dot{x}_1(0) + \dot{x}_2(0)\}^2}{\omega_1^2}\right]^{1/2}$$

$$X_1^{(2)} = \frac{1}{(r_2 - r_1)}\left[\{-r_1 x_1(0) + x_2(0)\}^2 + \frac{\{r_1\dot{x}_1(0) - \dot{x}_2(0)\}^2}{\omega_2^2}\right]^{1/2}$$

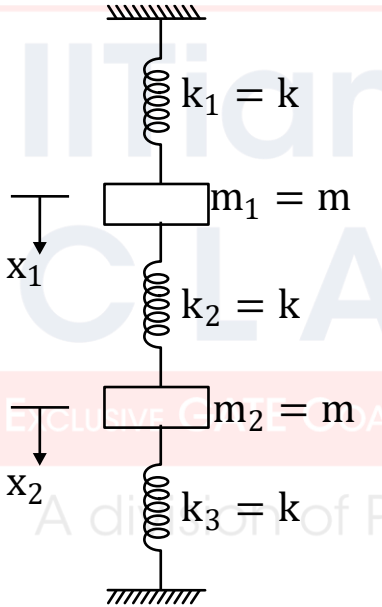
$$\phi_1 = tan^{-1}\left\{\frac{-r_2\dot{x}_1(0) + \dot{x}_2(0)}{\omega_1[r_2 x_1(0) - x_2(0)]}\right\}$$

$$\phi_2 = tan^{-1}\left\{\frac{r_1\dot{x}_1(0) - \dot{x}_2(0)}{\omega_2[-r_1 x_1(0) + x_2(0)]}\right\}$$

EXAMPLE PROBLEM

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Spring system mass m and spring stiffness k for individual mass and spring is given below in figure.



$x_2 > x_1$

kx_1 ↑ m ↑ $m\ddot{x}_1$ ↓ $k(x_2 - x_1)$

$kx_1 + m\ddot{x}_1 - k(x_2 - x_1) = 0$

$m\ddot{x}_1 + 2kx_1 - kx_2 = 0$

$k(x_2 - x_1)$ ↑ m ↑ $m\ddot{x}_2$ ↑ kx_2

$k(x_2 - x_1) + m\ddot{x}_2 + kx_2 = 0$

$m\ddot{x}_2 - kx_1 + 2kx_2 = 0$

Finding natural frequency & mode shape plot of spring mass system

EXAMPLE PROBLEM

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Equation of motion can be written

$$m\ddot{x}_1 + 2kx_1 - kx_2 = 0$$

$$m\ddot{x}_2 - kx_1 + 2kx_2 = 0$$

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} 2k & -k \\ -k & 2k \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -m\omega^2 + 2k & -k \\ -k & 2k - m\omega^2 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$x_1 = X_1 \cos(\omega t + \phi)$$

$$x_2 = X_2 \cos(\omega t + \phi)$$

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EXAMPLE PROBLEM- MODE SHAPE PLOT

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$$det\begin{bmatrix}(-m\omega^2+2k) & -k\\ -k & (-m\omega^2+2k)\end{bmatrix}=0$$

$m^2\omega^4 - 4km\omega^2 + 3k^2 = 0$

By solving,

$\omega_1 = \sqrt{k/m}$

$\omega_2 = \sqrt{3k/m}$

Let $\omega^2 = p$

$\Rightarrow m^2p - 4kmp + 3k^2 = 0$

$\Rightarrow p = k/m \;,\; p = 3k/m$

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EXAMPLE PROBLEM- MODE SHAPE PLOT

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$$\begin{bmatrix} -m\omega^2+2k & -k \\ \cdots & \cdots \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Amplitude Ratio,

(2)

$$r_1 = \frac{X_2^{(1)}}{X_1^{(1)}} = \frac{-m\omega_1^2 + 2k}{k} = \frac{k}{-m\omega_1^2 + 2k} = 1$$

$(-m\omega_1^2+2k)X_1^{(1)} - kX_2^{(1)} = 0$

$\frac{X_2^{(1)}}{X_1^{(1)}} = \frac{(-m\omega_1^2+2k)}{+k} = r_1 = 1$

On substituting $\omega_1 \Rightarrow r_1 = 1 \Rightarrow X_2^{(1)} = X_1^{(1)}$

$$r_2 = \frac{X_2^{(2)}}{X_1^{(2)}} = \frac{-m\omega_2^2 + 2k}{k} = \frac{k}{-m\omega_2^2 + 2k} = -1$$

$\Rightarrow r_2 = -1 \quad \Rightarrow X_2^{(2)} = -X_1^{(2)}$

EXAMPLE PROBLEM- MODE SHAPE PLOT

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The natural modes are given,

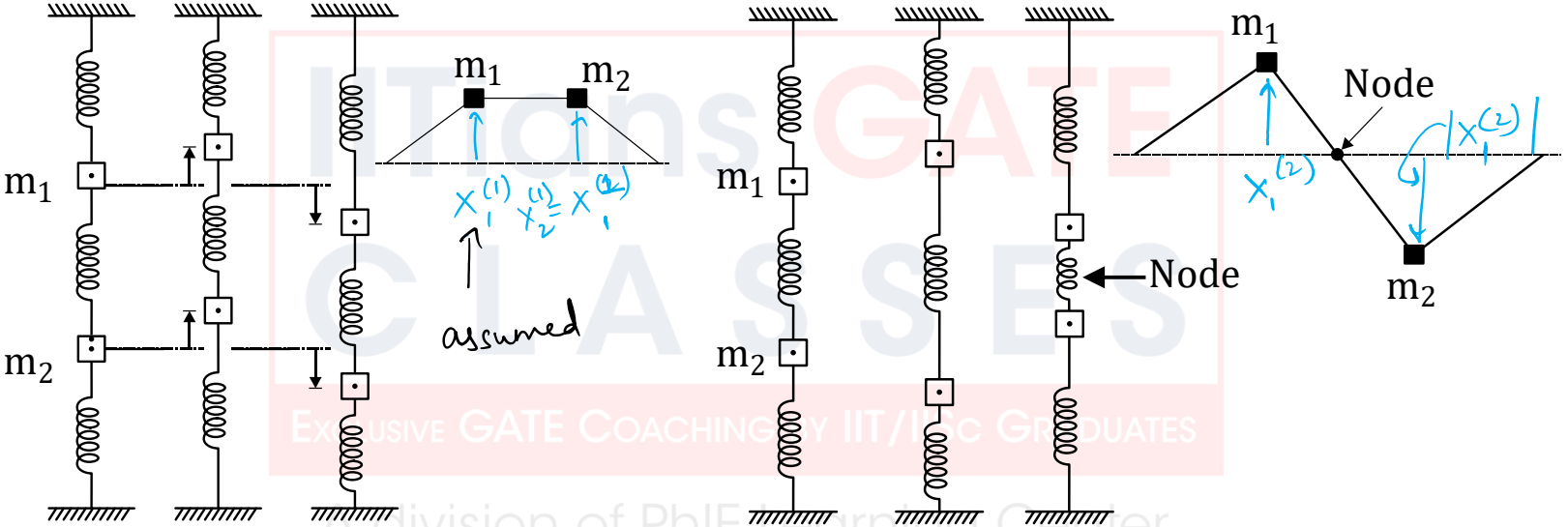
$$\text{First mode} = \vec{x}^{(1)}(t) = \begin{Bmatrix} X_1^{(1)} \cos\left(\sqrt{\frac{k}{m}}t + \phi_1\right) \\ X_2^{(1)} \cos\left(\sqrt{\frac{k}{m}}t + \phi_1\right) \end{Bmatrix}$$

$$\text{Second mode} = \vec{x}^{(2)}(t) = \begin{Bmatrix} X_1^{(2)} \cos\left(\sqrt{\frac{3k}{m}}t + \phi_2\right) \\ -X_2^{(2)} \cos\left(\sqrt{\frac{3k}{m}}t + \phi_2\right) \end{Bmatrix}$$

EXAMPLE PROBLEM- MODE SHAPE PLOT

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Mode shape – Displacement of each particle in system at a particular mode/frequency.



(a) First Mode (ω_1) $\quad X_2^{(1)} = X_1^{(1)}$

(b) Second Mode $\quad X_2^{(2)} = -X_1^{(2)}$

$\because$ for particle '1' displacement is assumed $\therefore$ Mode shapes are relative displacement

EXAMPLE PROBLEM- MODE SHAPE PLOT

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Equation of motion can be written,

$x_1^{(1)}(t) + x_1^{(2)}(t)$

$$x_1(t) = X_1^{(1)} \cos\left(\sqrt{\frac{k}{m}}t + \phi_1\right) + X_1^{(2)} \cos\left(\sqrt{\frac{3k}{m}}t + \phi_2\right)$$ ← Response of particle m_1

$$x_2(t) = X_2^{(1)} \cos\left(\sqrt{\frac{k}{m}}t + \phi_1\right) - X_2^{(2)} \cos\left(\sqrt{\frac{3k}{m}}t + \phi_2\right)$$ ← Response of particle m_2

$x_2^{(1)}(t) + x_2^{(2)}(t)$

$r_1 = 1$

$r_2 = -1$

PROBLEM 4

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If natural frequencies are given by

$\omega_{n1} = 30\, rad/sec;\ \ \omega_{n2} = 40\, rad/sec$

Mode shapes for this system have been shown to be

$$\left(\frac{X_1}{X_2}\right)_1 = 1, \qquad \left(\frac{X_1}{X_2}\right)_2 = -1$$

If both the masses are its respective equilibrium position without having any velocity, find the final response of first mass

$A)\ x_1(t) = X_1^{(1)} cos(40t + \phi_1) + X_1^{(2)} cos(30t + \phi_2)$

$B)\ x_1(t) = X_2^{(1)} cos(30t + \phi_1) + X_1^{(2)} cos(40t + \phi_2)$

$C)\ x_1(t) = X_2^{(1)} cos(30t + \phi_1) - X_2^{(2)} cos(40t + \phi_2)$

$D)\ x_1(t) = X_1^{(1)} cos(30t + \phi_1) + X_2^{(2)} cos(40t + \phi_2)$

Ans c

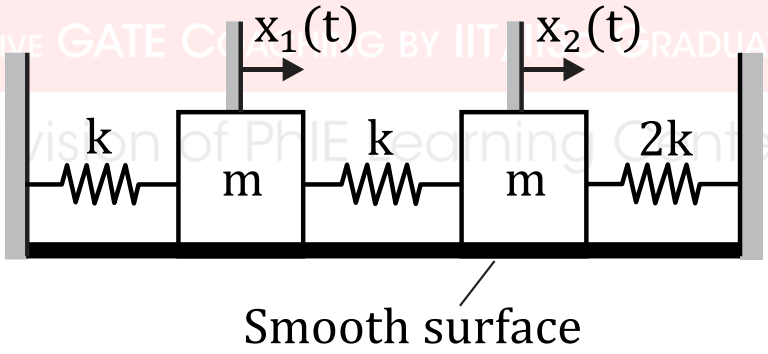
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PROBLEM 5

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The two masses of the two degree of freedom system shown in the figure are given initial displacements of 2 cm (x_1) and 1.24 cm (x_2). The system starts to vibrate in the first mode. The first mode shape of this system is $\phi_1 = [1 \ a]^T$, where a = ___ (round off to two decimal places).



Given:- $x_1(t) = 2$, $x_2(t) = 1.24$

$x_1(t) = x_1^{(1)}(t) + x_1^{(2)}(t)$

$x_2(t) = x_2^{(1)}(t) + x_2^{(2)}(t)$

Mode shape 1

Mode shape 2

$x_1(t) = x_1^{(1)}(t) = X_1' \cos(\omega_1 t + \phi)$

$x_2(t) = x_2^{(1)}(t) = X_2' \cos(\omega_1 t + \phi)$

$$\begin{Bmatrix} x_1(t) \\ x_2(t) \end{Bmatrix} = \begin{Bmatrix} X_1' \\ X_2' \end{Bmatrix}$$

$$\begin{Bmatrix} 2 \\ 1.24 \end{Bmatrix} = \begin{Bmatrix} X_1' \\ X_2' \end{Bmatrix} = \begin{bmatrix} 1 \\ a \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 1.24/2 \end{bmatrix} = \begin{bmatrix} 1 \\ a \end{bmatrix}$$

$\Rightarrow a = 0.62$

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